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SUPERSEDING
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2 October 1987

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE, PRECISION VOLTAGE REFERENCE, SHUNT REGULATOR, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, precision voltage reference. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN) (see 6.6).

1.2 Classification.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Adjustable voltage reference, shunt regulator
02	2.5 V voltage reference, shunt regulator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
P	D-4 (8-lead, .405" x .310" x .200") dual-in-line package
X	See figure 1, 3-lead can package

1.3 Absolute maximum ratings.

Cathode to anode voltage (device type 01)	37 V dc
Continuous cathode current range (device type 01)	-100 to +150 mA
Reference input current (device type 01)	-0.05 to 10 mA
Maximum power dissipation (P_D) ambient (device type 01)	1,050 mW 1/
Reverse current (device type 02)	20 mA
Forward current (device type 02)	10 mA
Storage temperature range	-65°C to +150°C
Junction operating temperature (T_J)	+150°C
Load temperature (soldering, 10 seconds)	+300°C

1/ Derate 8.4 mW/°C above T_A = +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5962

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1.4 Recommended operating conditions.

Cathode to anode voltage (device type 01)	36 V
Cathode current (device type 01)	100 mA
Reverse current (device type 02)	18 mA
Forward current (device type 02)	8 mA
Ambient operating temperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum θ_{JC}	Maximum θ_{JA}
8 pin DIP	P	38°C/W	100°C/W
3 pin CAN	X	80°C/W	440°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Functional diagram and terminal connections. The functional diagram and terminal connections shall be as specified on figure 2.

3.2.2 Functional schematic circuit. The functional schematic circuit shall be as specified on figure 3.

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TABLE I. Electrical performance characteristics for device type 01.

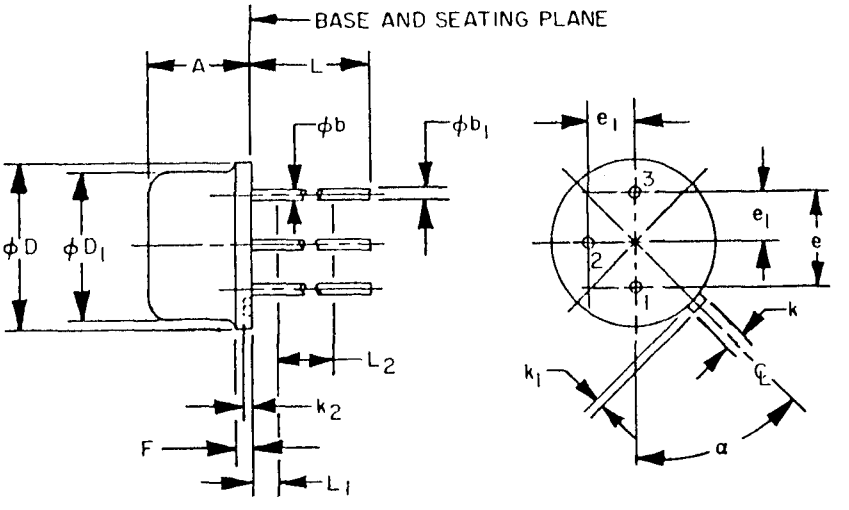
Characteristic	Symbol	Conditions R_1 = 10 kΩ, I_K = 10 mA (see 3.4) unless otherwise specified		Limits		Unit
				Min	Max	
Reference input	V_{REF}	$V_{KA} = V_{REF}$ see figure 4	T_A = -55°C, +125°C	2.42	2.57	V
			T_A = +25°C	2.44	2.55	
Cathode voltage	V_{KA10}	$V_{KA10} = V_{KA}$ R_2 = 3.33 kΩ, see figure 5		8.00	12.00	V
Cathode voltage	V_{KA36}	$V_{KA36} = V_{KA}$ R_2 = 746Ω, see figure 5		31.00	36.00	V
Ratio of change in V_{REF} to change in V_{KA}	$\frac{\Delta V_R}{\Delta V_{K(1)}}$	$V_{REF} = V_{R2} - V_{R1}$ $V_{KA} = V_{KA10} - V_{R1}$ R_2 = 3.33 kΩ, see figure 5			-2.70	mV/V
Ratio of change in V_{REF} to change in V_{KA}	$\frac{\Delta V_R}{\Delta V_{K(2)}}$	$V_{REF} = V_{R3} - V_{R1}$ $V_{KA} = V_{KA36} - V_{R1}$ R_2 = 746Ω, see figure 5			-2.00	mV/V
Reference input current	I_{REF}	$I_R = I_{REF}$ see figure 5	T_A = +25°C	-0.1	4.00	µA
			T_A = -55°C, +125°C	-0.1	7.00	µA
Minimum cathode current for regulation	I_{MIN}	$V_{KA} = V_{REF}$, I_K = 1 mA R_2 = ∞, see figure 4		2.40	2.60	V
Off-state cathode	I_{OFF}	V_{KA} = 36 V, V_{REF} = 0 V see figure 6		-0.1	1.00	µA
Input impedance	Z_{KA}	$V_{KA} = V_{REF}$, I_K = 1.0 mA to 100 mA, see figure 4			0.50	Ω
Noise	N_O	I_K = 10 mA B_W = 0.1 Hz to 10 Hz see figure 7, T_A = +25°C			20.00	µVp-p

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TABLE I. Electrical performance characteristics for device type 02.

Characteristic	Symbol	Conditions V_{REF} = 15 V, -55°C < T_A < +125°C unless otherwise specified	Limits Min	Limits Max	Unit
Minimum operating current	I_{MIN}	V_Z = 2.48 V, T_A = +25°C		260	µA
		V_Z = 2.465 V, T_A = +125°C		340	
		V_Z = 2.465 V, T_A = -55°C		200	
Voltage reference	V_Z	I_R = 400 µA T_A = +25°C	2.495	2.505	V
		I_R = 400 µA T_A = -55°C, +125°C	2.480	2.520	
		I_R = 1 mA T_A = +25°C	2.495	2.505	
		I_R = 1 mA T_A = -55°C, +125°C	2.480	2.520	
		I_R = 10 mA T_A = +25°C	2.490	2.510	
		I_R = 10 mA T_A = -55°C, +125°C	2.480	2.520	
Load regulation	$V_Z$load	I_R = 400 µA, 10 mA	-10.0	10.0	mV
Voltage adjust reference	V_{ADJ}	ADJ = 600 mV	2.500	2.500	V
		ADJ = V_{REF} - 600 mV			
Reference adjustment	V_ZADJ		150.0		mV
Impedence	V_Z/I_Z	400 uA < I_Z < 10 mA		1.0	Ohms
Delta voltage/ delta temperature	$\frac{V_Z}{dT}$	From -55°C to +25°C		10	mV
		From +25°C to +125°C		10	
		I_R = 1 mA			
Noise	N_O	I_R = 1 mA, BW = 0.1 Hz to 10 Hz		30	µVp-p

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	0.085	0.105	2.159	2.667	
øb	0.016	0.019	0.406	0.483	3
øb1					3
øD	0.209	0.219	5.300	5.503	
øD1	0.178	0.195	4.521	4.953	

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
e	0.100 T.P.		2.540 T.P.		5
e1	0.050 T.P.		1.270 T.P.		5
F	---	0.030	---	0.762	
k	0.036	0.046	0.914	1.168	
k1	0.028	0.048	0.711	1.219	4

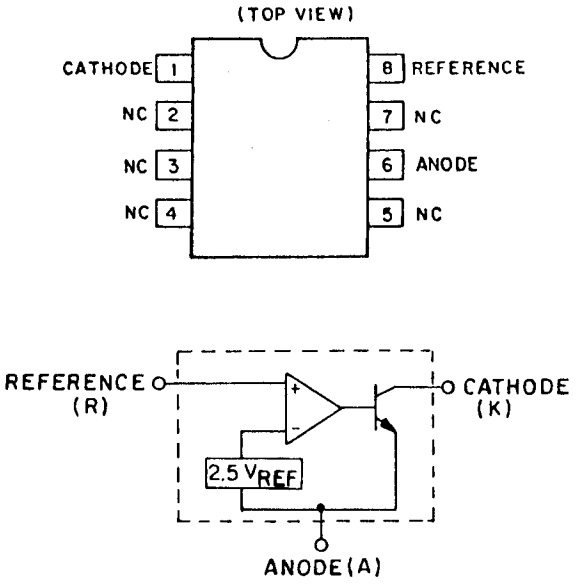
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
k2					
L	0.500	--	12.700	--	
L1	---	0.50	---	1.27	
L2	.250	--	6.35	--	
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. (Three leads) øb applies between L1 and L2. øb1 applies between L2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L1 and beyond .500 912.70 mm) from the reference plane.
4. Three leads.
5. Measured from the maximum diameter of the product.
6. Leads having a maximum diameter .019 (0.48 mm) measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the base plane of the product shall be within .007 (0.18 mm) of their true position relative to a maximum width tab.
7. The product may be measured by direct methods or by gauge.

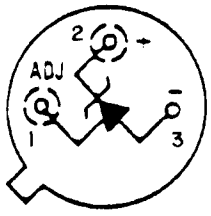
FIGURE 1. Case outline X (device type 02).

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Device type 01



Device type 02



(Bottom view)

FIGURE 2. Functional diagram and terminal connections.

Device type 01

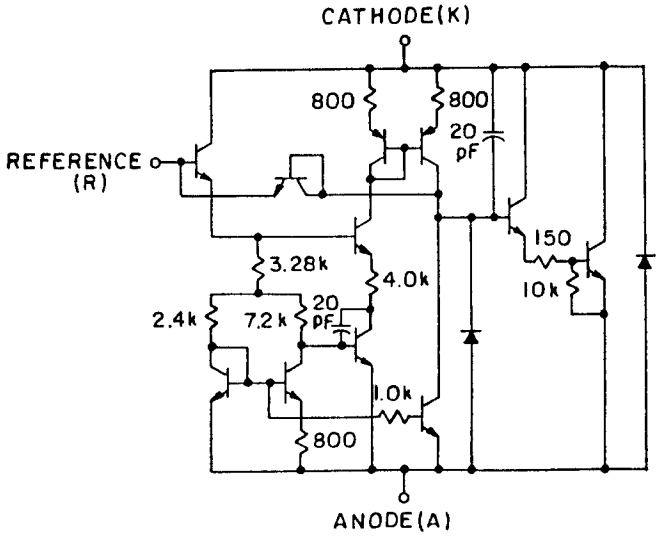


FIGURE 3. Functional schematic circuit.

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Device type 02

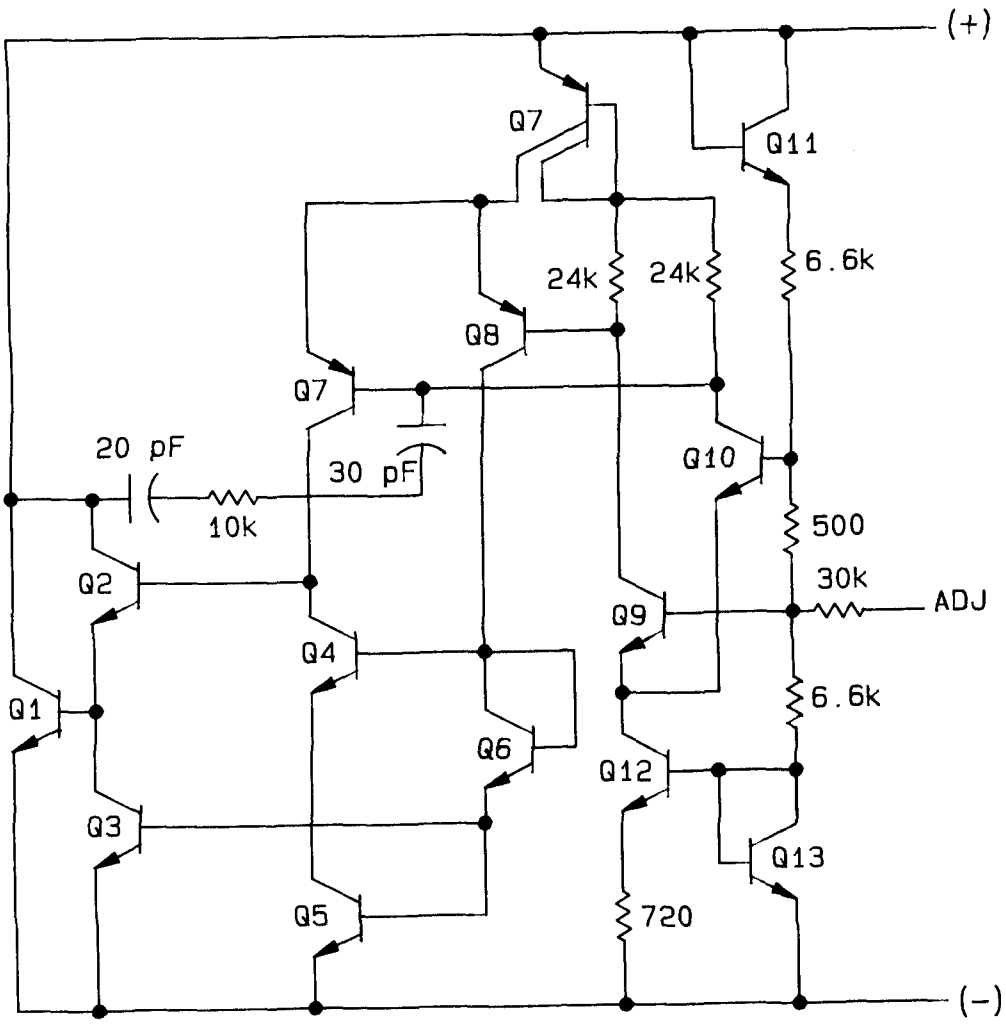


FIGURE 3. Functional schematic circuit - Continued.

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Device type 01

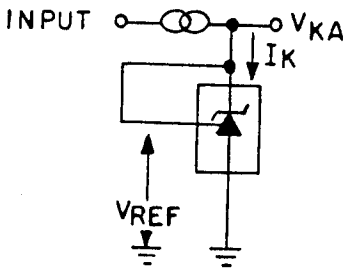


FIGURE 4. Test circuit for ($V_{KA} = V_{REF}$).

Device type 01

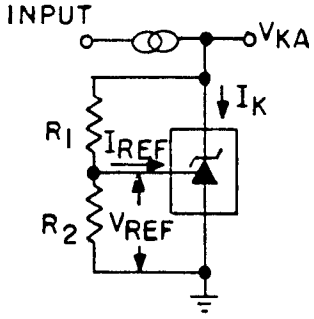


FIGURE 5. Test circuit for ($V_{KA} > V_{REF}$).

Device type 01

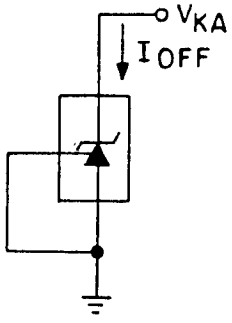


FIGURE 6. Test circuit for I_{OFF}.

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Device type 02

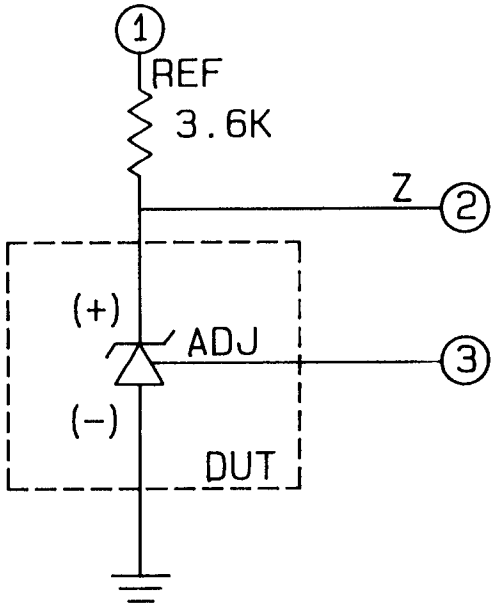
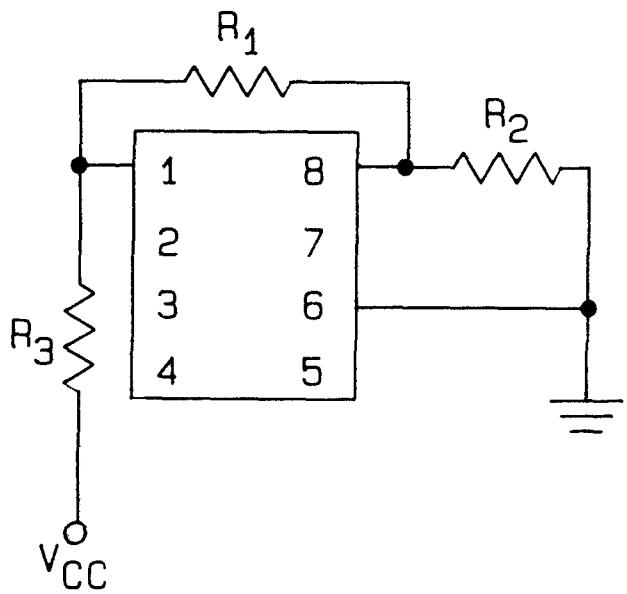


FIGURE 6. Test circuit - Continued.

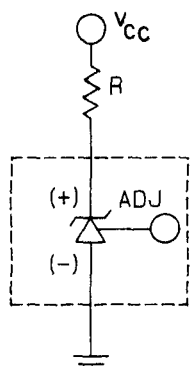
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Device type 01



NOTES:
1. R_1 = 21.5 K ±1 percent, .250 w-metal film.
2. R_2 = 1.78 K ±1 percent, .250 w-metal film.
3. R_3 = 1.27 K ±1 percent, .250 w-metal film.
4. V_{CC} = 40 V +0 V
   -3 V

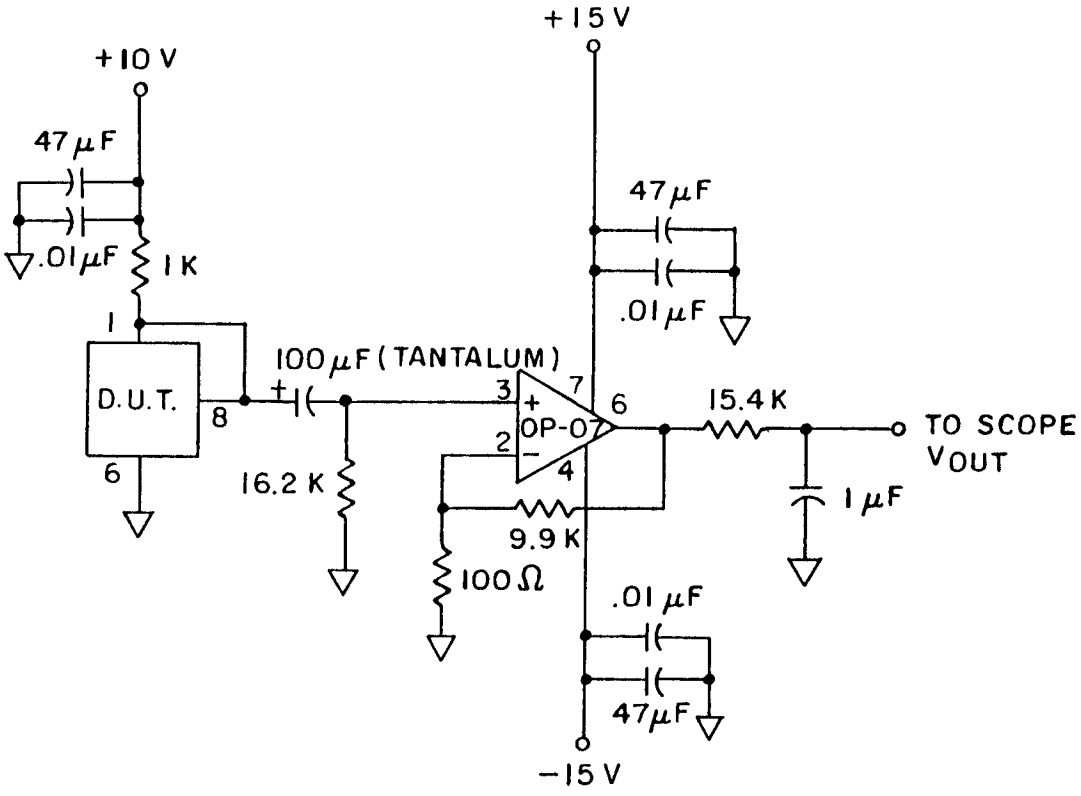
Device type 02



1. R = 2 kΩ ±1 percent, 0.250 w-metal film.
2. V_{CC} = +20 V +0 V
   -1 V

FIGURE 7. Burn- in and steady -state life test circuit.

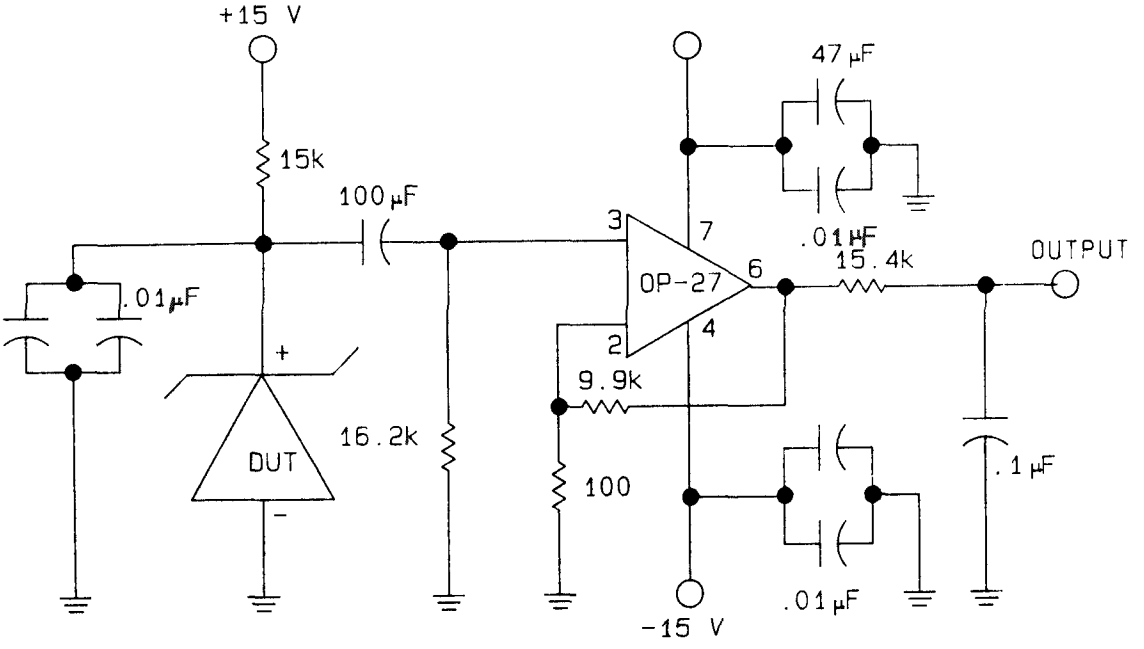
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NOTES:
1. Test time = 10 ns.
2. V_{OUT} measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.

FIGURE 8. Low frequency noise test circuit for device type 01.

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NOTES:
1. Test time = 10 ns.
2. V_{OUT} measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.

FIGURE 8. Noise test circuit for device type 02 - Continued.

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3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and available upon request.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full ambient operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by PIN. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 59 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1, 2, 3	1, 2, 3
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6	1, 2, 3, 4, 5, 6
Group C end-point and group B, class S electrical parameters (method 5005)	1, 2, 3, and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1, 2, 3	1
Additional electrical subgroups for group C periodic inspection	N/A	---

1/ *PDA applies to subgroup 1 (see 4.2c).

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Conditions R_1 = 10 kΩ, I_K = 10 mA unless otherwise specified	Measured value	See figure	Notes	Limits		Unit
							Min	Max	
1 T_A = +25°C	V_{REF1}	1	$V_{KA} = V_{REF}$	V_{R1}	3	$V_R = V_{EF}$	2.44	2.55	V dc
	V_{KA10}	2	R_2 = 3.33 kΩ	V_{KA10}	4	$V_{KA10} = V_{KA}$ 1/	8.00	12.00	V dc
	V_{KA36}	3	R_2 = 746Ω	V_{KA36}	4	$V_{KA36} = V_{KA}$ 1/	31.00	36.00	V dc
	$\Delta V_R/\Delta V_K$	4	R_2 = 3.33 kΩ	V_{R2}	4	$\Delta V_{REF} = V_{R2} - V_{R1}$ $\Delta V_{KA} = V_{KA10} - V_{R1}$		-2.70	mV/V
		5	R_2 = 746Ω	V_{R3}	4	$\Delta V_{REF} = V_{R3} - V_{R2}$ $\Delta V_{KA} = V_{KA36} - V_{KA10}$		-2.00	mV/V
	I_{REF}	6	$R_2 = \infty$	I_{R1}	4	$I_R = I_{REF}$	-0.1	4.00	µA
	I_{MIN}	7	$V_{KA} = V_{REF}$ I_{KMIN} = 1 mA	V_{R4}	3	I_K = Cathode current	2.40	2.60	V dc
	I_{OFF}	8	V_{KA} = 36 V, V_{REF} = 0 V	I_{K1}	5		-0.1	1.00	µA
	Z_{KA}	9	$V_{KA} = V_{REF}$, I_K = 100 mA	V_{R5}	3	$Z_{KA} = (V_{R5}-V_{R4})/99$ mA 2/		0.50	Ω
2 T_A = +125°C	V_{REF1}	10	$V_{KA} = V_{REF}$	V_{R6}	3	$V_R = V_{REF}$	2.42	2.57	V dc
	V_{KA10}	11	R_2 = 3.33 kΩ	V_{KA10}	4	$V_{KA10} = V_{KA}$ 1/	8.00	12.00	V dc
	V_{KA36}	12	R_2 = 746Ω	V_{KA36}	4	$V_{KA36} = V_{KA}$ 1/	31.00	36.00	V dc
	$\Delta V_R/\Delta V_K$	13	R_2 = 3.33 kΩ	V_{R7}	4	$\Delta V_{REF} = V_{R7} - V_{R6}$ $\Delta V_{KA} = V_{KA10} - V_{R6}$		-2.70	mV/V
		14	R_2 = 746Ω	V_{R8}	4	$\Delta V_{REF} = V_{R8} - V_{R7}$ $\Delta V_{KA} = V_{KA36} - V_{KA10}$		-2.00	mV/V
	I_{REF}	15	$R_2 = \infty$	I_{R2}	4	$I_R = I_{REF}$	-0.1	7.00	µA
	I_{MIN}	16	$V_{KA} = V_{REF}$ I_{KMIN} = 1 mA	V_{R9}	3	I_K = Cathode current	2.40	2.60	V dc
	I_{OFF}	17	V_{KA} = 36 V, V_{REF} = 0 V	I_{K2}	5		-0.1	1.00	µA
	Z_{KA}	18	$V_{KA} = V_{REF}$, I_K = 100 mA	V_{R10}	3	$Z_{KA} = (V_{R10}-V_{R6})/99$ mA 2/		0.50	Ω

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions R_1 = 10 kΩ, I_K = 10 mA unless otherwise specified	Measured value	See figure	Notes	Limits		Unit
							Min	Max	
3 T_A = -55°C	V_{REF1}	19	$V_{KA} = V_{REF}$	V_{R11}	3	$V_R = V_{REF}$	2.42	2.57	V dc
	V_{KA10}	20	R_2 = 3.33 kΩ	V_{KA10}	4	$V_{KA10} = V_{KA}$ 1/	8.00	12.00	V dc
	V_{KA36}	21	R_2 = 746Ω	V_{KA36}	4	$V_{KA36} = V_{KA}$ 1/	31.00	36.00	V dc
	$\Delta V_R/\Delta V_K$	22	R_2 = 3.33 kΩ	V_{R12}	4	$\Delta V_{REF} = V_{R12} - V_{R11}$ $\Delta V_{KA} = V_{KA10} - V_{R11}$		-2.70	mV/V
		23	R_2 = 746Ω	V_{R13}	4	$\Delta V_{REF} = V_{R13} - V_{R12}$ $\Delta V_{KA} = V_{KA36} - V_{KA10}$		-2.00	mV/V
	I_{REF}	24	$R_2 = \infty$	I_{R3}	4	$I_R = I_{REF}$	-0.1	7.00	µA
	I_{MIN}	25	$V_{KA} = V_{REF}$ I_{KMIN} = 1 mA	V_{R14}	3	I_K = Cathode current	2.40	2.60	V dc
	I_{OFF}	26	V_{KA} = 36 V, V_{REF} = 0 V	I_{K3}	5		-0.1	1.00	µA
	Z_{KA}	27	$V_{KA} = V_{REF}$, I_K = 100 mA	V_{R15}	3	$Z_{KA} = (V_{R15}-V_{R14})/99$ mA 2/		0.50	Ω
4 T_A = +25°C	N_O	28	I_K = 10 mA, B_W = 0.1 Hz to 10 Hz	N_O	6			20.00	µVp-p
5 T_A = +125°C	$\Delta V_{REF}/\Delta T$	29	$V_{KA} = V_{REF}$	CALC	3	$\Delta V_{REF} = V_{R1} - V_{R6}$	-44.00	44.00	mV
6 T_A = -55°C	$\Delta V_{REF}/\Delta T$	30	$V_{KA} = V_{REF}$	CALC	3	$\Delta V_{REF} = V_{R1} - V_{R11}$	-44.00	44.00	mV
	$\Delta V_{REF}/\Delta T$	31	$V_{KA} = V_{REF}$	CALC	3	$\Delta V_{REF} = V_{R11} - V_{R6}$	-44.00	44.00	mV
	$\Delta I_{REF}/\Delta T$	32	$R_2 = \infty$	CALC	4	$\Delta I_{REF} = I_{R2} - I_{R3}$	-3.00	3.00	µA

1/ $V_{KA} = V_{REF}(1 + R_1/R_2) + I_{REF} \times R_2$ (see figure 4).

2/ $Z_{KA} = \Delta V_{KA}/\Delta I_K$, f < 1 kHz.

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TABLE III. Group A inspection for device 02. 1/

Subgroup	Symbol	Test no.	Measured value	Reference	Z	Adjustment	Pin measurement	Equations	Limits		Units
									Min	Max	
1 T_A = +25°C	I_{MIN}	1	I1	260 µA			1		2.48		V
	V_Z	2	V1	400 µA			2	V_Z = V1	2.495	2.505	V
		3	V2	1 mA			2	V_Z = V2	2.495	2.505	
		4	V3	10 mA			2	V_Z = V3	2.490	2.510	
	V_{ZLOAD}	5						V_{ZLOAD} = V1 - V3	-10.0	10.0	mV
	V_{ADJ}	6	V4			600 mV	2	V_{ADJ} = V4		2.500	V
		7	V5			1.9 V	2	V_{ADJ} = V5	2.500		
	V_{ZADJ}	8					2	V_{ZADJ} = V5 - V4	150.0		mV
	V_Z/I_Z	9	V6	1 mA			2	Z_{IN} = (V6 - V7)/9.6 mA		1.0	Ω
		10	V7	10 mA			2				
2 T_A = +125°C	I_{MIN}	11	I2	340 µA			1		2.465		V
	V_Z	12	V8	400 µA			2	V_Z = V8	2.480	2.520	V
		13	V9	1 mA			2	V_Z = V9	2.480	2.520	
		14	V10	10 mA			2	V_Z = V10	2.480	2.520	
	V_{ZLOAD}	15						V_{ZLOAD} = V10 - V8	-10.0	10.0	mV
	V_{ADJ}	16	V11			600 mA	2	V_{ADJ} = V11		2.500	V
		17	V12			1.9 V	2	V_{ADJ} = V12	2.500		
	V_{ZADJ}	18					2	V_{ZADJ} = V12 - V11	150.0		mV
	V_Z/I_Z	19	V13	1 mA			2	Z_{IN} = (V14 - V13)/9.6 mA		1.0	Ω
		20	V14	10 mA			2				

See footnote at end of table.

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TABLE III. Group A inspection for device 02 - Continued. 1/

Subgroup	Symbol	Test no.	Measured value	Reference	Z	Adjustment	Pin measurement	Equations	Limits		Units
									Min	Max	
3 T_A = -55°C	I_{MIN}	21	I3	200 µA			1		2.465		V
	V_Z	22	V15	400 µA			2	V_Z = V15	2.480	2.520	V
		23	V16	1 mA			2	V_Z = V16	2.480	2.520	
		24	V17	10 mA			2	V_Z = V17	2.480	2.520	
	V_{ZLOAD}	25						V_{ZLOAD} = V15 - V17	-10.0	10.0	mV
	V_{ADJ}	26	V18			600 mV	2	V_{ADJ} = V18		2.500	V
		27	V19			1.9 V	2	V_{ADJ} = V19	2.500		
	V_{ZADJ}	28					2	V_{ZADJ} = V19 - V18	150.0		mV
	V_Z/I_Z	29	V20	1 mA			2	Z_{IN} = (V21 - V20)/9.6 mA		1.0	Ω
		30	V21	10 mA			2				
4 T_A = +25°C	N_O	31	N1					See figure 7		30	µVp-p
5 T_A = +125°C	$\frac{\Delta V_{OUT}}{\Delta T}$	32						$\frac{\Delta V_{OUT}}{\Delta T}$ = V9 - V2		10	mV
6 T_A = -55°C	$\frac{\Delta V_{OUT}}{\Delta T}$	33						$\frac{\Delta V_{OUT}}{\Delta T}$ = V16 - V2		10	mV

1/ Use test circuit for device type 02 shown on figure 6.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein. When the accelerated test conditions (T_A = +200°C or +250°C) are used for any test on a given inspection lot, they shall be used for all pertinent product assurance tests (screening (4.2), and group C inspection (4.4.3)) made on the lot.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) For class S devices, test condition D, using the circuit shown on figure 7.

(2) For class B devices, test condition D, using the circuit shown on figure 7.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 7, 8, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using a circuit shown on figure 7. If the alternate burn-in conditions are used, the circuit shown on figure 7 shall be used.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 7.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

TABLE IV. Group C end-point electrical parameters, T_A = +25°C.

Symbol	Delta 1/		End-point limits		Unit	Device type
	Min	Max	Min	Max		
V_{REF}	-0.015	0.015	2.44	2.55	V	01
I_{REF}	-1.0	1.0	-0.1	3.0	µA	
V_Z for I_R = 1 mA			2.495	2.505	V	02

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltage values are referenced to the external zero reference level of the supply voltage. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled within 10°C of their power stable condition at room temperature prior to removal of the bias.

4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be applied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2c).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete Part or Identifying Number (PIN) (see 6.6).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. *Requirements for certificate of compliance, if applicable.*

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

h. Requirements for product assurance options.

i. *Requirements for special lead lengths, or lead forming, if applicable. These requirements* shall not affect the PIN.

j. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331 (including terms and symbols for device terminals), and as follows:

V_{REF}	-	Reference input voltage
V_{KA10}	-	Cathode voltage
V_{KA36}	-	Cathode voltage
$V_R/V_K(1)$	-	Ratio of change in V_{REF} to change in V_{KA}
$V_R/V_K(2)$	-	Ratio of change in V_{REF} to change in V_{KA}
I_{REF}	-	Reference input current
I_{MIN}	-	Minimum cathode current for regulation
I_{OFF}	-	Off-state cathode
Z_{KA}	-	Input impedence
I_{MIN}	-	Minimum operating current
V_Z	-	Voltage reference
V_{ZLOAD}	-	Load regulation
V_{ADJ}	-	Voltage adjust reference
V_{ZADJ}	-	Reference adjustment
V_Z/I_Z	-	Impedence
V_Z/d_T	-	Delta voltage/Delta temperature
N_o	-	Noise

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6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the PIN.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	TL431
02	LT1009

6.6 Part or Identifying Number (PIN). The PIN is created as specified in MIL-M-38510.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, AV
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1184)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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